

Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

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Abstract

Completion-audit workflows often stop before the remaining workload is known. Their stop signals—no-new findings, agent agreement, stable summaries, or self-reported completion—can be locally meaningful while still being unsafe as scope-wide completion certificates. The missing question is: under what evidence condition was the stop signal produced, and does that condition match the claimed scope? We study this certificate mismatch in bounded source-route audit workflows. A source-route stratum records both where the workflow searched and which audit route it applied. We use the resulting exposure geometry to build an evidence-condition controller with four modules: source-route exposure estimation, an eligibility gate, residual-evidence repair over weak plausible gaps, and a final SAFE/CONTINUE/ABSTAIN decision rule. Across four bounded completion-audit task families, homogeneous route reuse creates false certification risk; source-route granularity detects this risk where source-only control does not; broad eligibility alone is insufficient on the `urllib3` boundary; and the full controller avoids unsafe stops without becoming a never-stop rule. The result is a bounded diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

1 Introduction

Completion-audit workflows must eventually stop, even when the remaining workload is unknown. A local stop signal—no new findings, agreement, or a stable summary—can be true under the evidence condition that produced it while still failing as a certificate for the declared task scope. We call this *certificate mismatch*. In a code audit, for example, exhausting timeout evidence in a file does not certify TLS, retry, exception, or cleanup evidence in that same file.

We model this mismatch with *evidence-condition geometry*. A source-route stratum records both where a workflow searched and which audit route it applied. The exposure distribution over these strata is a runtime certificate diagnostic: it tells us whether a proposed stop is supported by broad source-route evidence or by a localized condition. The diagnostic does not know the hidden workload and does not prove open-world completion.

We build a controller around this diagnostic. It estimates source-route exposure, checks eligibility, repairs weak plausi-

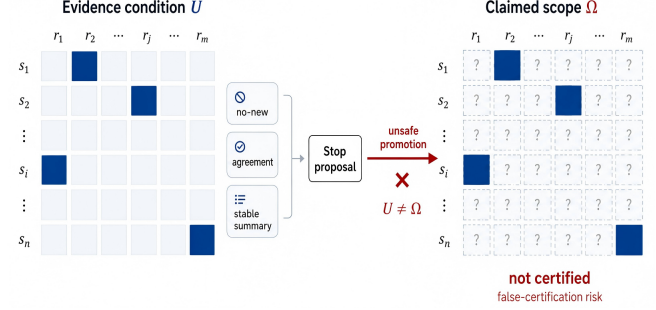


Figure 1: Certificate mismatch from local stop evidence. A workflow may observe no-new findings, agreement, or a stable summary under evidence condition U . Promoting that stop proposal to a completion claim over Ω is unsafe when $U \neq \Omega$; the unobserved source-route strata are not certified.

ble strata, and returns SAFE/CONTINUE/ABSTAIN. A generic verifier gate can avoid unsafe SAFE decisions by failing closed, but it gives no source-route diagnosis or repair target. Our controller aims for a stronger operational property: safe decisions when the condition is defensible, actionable CONTINUE decisions when residual evidence is found, and explicit ABSTAIN when no certificate can be formed. The contribution is therefore not a new retrieval policy alone; it is a certificate interface for deciding whether a stop proposal is safe, actionable, or still unsupported.

Experiments on two generated audits and two frozen real-repository audits test this safety–actionability–cost tradeoff. Homogeneous route reuse creates false certification risk; source-route geometry exposes risks hidden by source-only coverage; the `urllib3` boundary shows that eligibility is not proof; and the full controller avoids unsafe SAFE decisions while turning residual-positive unsafe states into actionable CONTINUE rather than merely abstaining. The paper makes three claims. First, stop evidence has a condition, and local conditions cannot certify broader source-route scopes. Second, source-route geometry is a useful stop-time warning signal because it separates file coverage from route coverage. Third, a repair-aware controller is more useful than a generic fail-closed verifier because it preserves safety while producing the next audit target when residual evidence appears.

The claim is bounded: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

2 Related Work

Our work sits between three literatures that usually treat stopping, evidence, and verification separately. High-recall review, total recall, active search, and review stopping rules ask how to retrieve more positives, estimate remaining mass, or decide whether a review has saturated under budget [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2015, 2016, Grossman et al., 2016, Roegiest et al., 2015, Voorhees, 2002, Wallace et al., 2010, Lewis and Gale, 1994, Settles, 2009, Gannett et al., 2012, Attenberg et al., 2015, Good, 1953, Chao, 1984, Chao et al., 2014]. Those methods are closest in spirit to our concern with false stopping, but they often reason with review protocols, sampling assumptions, prevalence estimates, or post-hoc recall targets. Our controller is deliberately poorer at runtime: it is not given oracle totals, hidden missing mass, post-hoc recall, or unseen true-item counts. It therefore asks a narrower certificate question: whether the observed evidence condition is broad enough to support the declared source-route scope of the completion claim.

A second line of work studies tool-using, web, code, and multi-agent systems that allocate work across searches, tools, memories, repositories, and intermediate states [Yao et al., 2023, Schick et al., 2023, Nakano et al., 2021, Qin et al., 2024, Wu et al., 2023, Park et al., 2023, Wang et al., 2024, Deng et al., 2024, Zhou et al., 2024, Liu et al., 2024, Yao et al., 2022, Jimenez et al., 2024, Shinn et al., 2024, Madaan et al., 2023]. These systems make our failure mode more common: a workflow may cover many sources but only through a narrow route, or may repeat one local route until the transcript appears stable. Benchmarks and factuality systems then evaluate final task success, answer support, or post-hoc correctness [Hendrycks et al., 2021a, Liang et al., 2022, Srivastava et al., 2023, Lin et al., 2022, Thorne et al., 2018, Gao et al., 2023, Min et al., 2023, Schuster et al., 2021, Gehrmann et al., 2021]. We instead evaluate the stop certificate itself: what condition produced it, and what action should follow when that condition is weak.

Finally, verification and oversight methods can reject unsupported outputs or fail closed when a checker reports unresolved warnings [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023, Amodei et al., 2016, Christiano et al., 2017, Irving et al., 2018, Leike et al., 2018, Hendrycks et al., 2021b]. We include a generic verifier-gate for exactly this reason. Its limitation is not that abstention is unsafe; abstention is often the right conservative answer. Its limitation is that a warning-only gate does not say which source-route condition is missing or where the next audit should go. Evidence-condition control treats the stop claim as a structured certificate: diagnose the source-route geometry, repair weak plausible strata, and return SAFE, CONTINUE, or ABSTAIN with an operational meaning.

3 Problem Setting

We study bounded completion-audit workflows with unknown residual workload and declared source-route scope. A workflow receives a task, allocates work across agents or tools, gath-

ers evidence under a finite budget, and must decide whether the task is complete within that declared scope. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. Every evidence event is produced under a source-route condition. A source-route stratum is $s = (x, r)$, and the intended evidence-condition space is

$$\Omega = S \times R.$$

The source-route space does not require the workflow to know all true missing items in advance. It requires the intended audit scope to be defined: which sources and routes are relevant to the completion claim. This assumption is weaker than knowing the answer set and stronger than an unbounded open-world search. It matches many practical audits: a system may know which repositories, documents, modules, or evidence sources are in scope, and it may know which audit routes are relevant, while still not knowing how many true findings remain. The controller therefore evaluates the condition under which evidence was gathered, not the unknown total workload itself.

A stop claim is a workflow assertion

$$C_t : T \text{ is complete over the intended workload scope.}$$

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the evaluation oracle:

$$\text{SAFE}(C_t) = \text{true}, \quad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, this is operationalized as accepting SAFE while bounded-oracle recall is below the completion threshold. In broader completion audits, it means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition. The oracle is used only for evaluation. It is not available to route assignment, repair scoring, eligibility checks, or stop decisions.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t . Exposure may count visits, scans, route-specific audits, tool calls, or other logged search actions. Define

$$p_{\text{exp}}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes where and how completion evidence was produced.

We use two runtime summaries:

$$\text{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\text{exp}}(t) = \text{Gini}(\{v_t(s) : s \in \Omega\}).$$

These summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

Proposition 1 (Local no-new evidence is not a global certificate). *Let $U \subset \Omega$ be a strict subset of the intended source-route space. Suppose all evidence used to support a stop claim C_t is produced inside U , and suppose the workflow observes no-new evidence only for actions inside U . Then the no-new evidence can support local exhaustion over U , but it cannot by itself support global completion over Ω .*

Proof sketch. No-new evidence is conditioned on the source-route actions that were executed. If no events are produced in $\Omega \setminus U$, the workflow has not observed the absence of residual evidence in those unexposed strata. A world in which U is exhausted and $\Omega \setminus U$ contains residual evidence is observationally indistinguishable from a globally complete world under the workflow’s local evidence log. Therefore local no-new evidence rules out additional findings only under the searched condition U ; without additional assumptions about unsearched strata, it cannot rule out residual evidence in $\Omega \setminus U$.

Source-only geometry is too coarse: auditing a file for time-out behavior does not certify that TLS, retry, exception, or cleanup behavior has been audited in that file. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and current experiments do not show a stable advantage over source-route. Source-route is the default because it captures both where and how evidence was produced while remaining interpretable and runtime-computable. The geometry is thus a certificate diagnostic rather than a semantic theory of all possible evidence. It asks whether the current log covers the intended source-route space broadly enough for the proposed stop claim; it does not assert that all routes are equally important or that support/Gini thresholds transfer unchanged across domains.

5 Method

The controller has four modules, shown in Algorithm 1. *Exposure estimation* maps the runtime evidence log to source-route counts $v_t(s)$, support $\text{support}(t)$, and Gini concentration $G_{\text{exp}}(t)$. *Eligibility* checks whether the stop condition is broad enough to consider a certificate. *Residual repair* probes weak plausible strata using only runtime-visible signals. The final rule returns SAFE only when the evidence condition is eligible and repair is nonproductive; residual evidence yields

Algorithm 1 Evidence-Condition Controller

Require: $\log E_t$, space Ω , claim C_t , budget B , thresholds τ_s, τ_g , repair rule $\mathcal{R}$

Ensure: decision in $\{\text{SAFE}, \text{CONTINUE}, \text{ABSTAIN}\}$

- 1: Compute $v_t(s)$, $p_{\text{exp}}(t, s)$, $\text{support}(t)$, and $G_{\text{exp}}(t)$
 - 2: $\text{eligible} \leftarrow [\text{support}(t) \geq \tau_s \wedge G_{\text{exp}}(t) \leq \tau_g]$
 - 3: **if** not *eligible* and $B = 0$ **then**
 - 4: **return** ABSTAIN
 - 5: **end if**
 - 6: **if** not *eligible* or weak plausible gaps remain **then**
 - 7: Audit $Q = \mathcal{R}(E_t, \Omega, C_t)$ within B
 - 8: **if** residual evidence appears **then**
 - 9: **return** CONTINUE
 - 10: **end if**
 - 11: Recompute *eligible*
 - 12: **end if**
 - 13: **if** *eligible* and no residual evidence appeared **then**
 - 14: **return** SAFE
 - 15: **end if**
 - 16: **return** ABSTAIN
-

actionable CONTINUE, and unresolved ineligibility yields ABSTAIN.

The repair instance used in experiments ranks strata by

$$\text{priority}(s) = \text{under_exposure}(s) \cdot \text{runtime_potential}(s).$$

The potential term may use source text, route names, route match counts, source size, or ledger-visible notes. It may not use oracle labels, oracle totals, post-hoc recall, hidden missing mass, or undiscovered true-item counts. Residual-potential is a mechanism-aligned probe, not an optimal active-search claim. Thus the method’s value is stop-time certificate diagnosis plus targeted repair, not universal completion proof.

Proposition 2 (Controller certificate semantics). *For a declared source-route scope Ω , thresholds τ_s, τ_g , and a fixed repair rule $\mathcal{R}$ that uses only runtime-visible signals, Algorithm 1 returns SAFE only if (i) the exposure condition satisfies the eligibility gate and (ii) repair or audit reveals no residual evidence within the allocated repair budget. If residual evidence is found, the returned decision is CONTINUE; if eligibility cannot be established under budget, the returned decision is ABSTAIN.*

The proof sketch is in the supplement; the proposition is about the controller’s declared semantics, not about absence of all unobserved evidence.

6 Experiments and Results

We evaluate whether a stop controller is *safe*, *actionable*, and *cost-aware* under a declared source-route scope. Each comparison fixes the workflow trajectory before scoring: a policy sees only runtime-visible evidence, optional repair scans a configured budget, and oracle labels are revealed only after the decision. Thus the experiments test completion con-

Evidence-Condition Controller

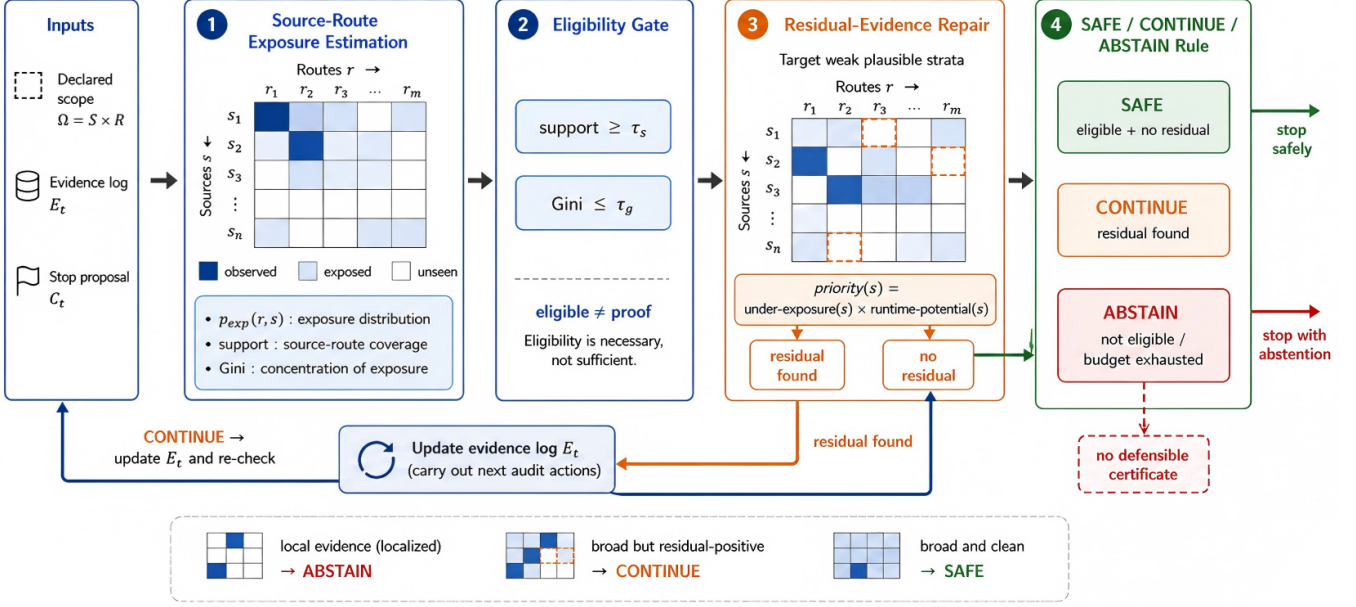


Figure 2: Evidence-condition controller. The controller maps a runtime evidence log, declared source-route scope, and stop proposal to an exposure distribution over the intended source-route space. The support/Gini check is an eligibility test, not a completion proof. If the condition is localized or weak plausible gaps remain, the controller repairs or continues. It returns SAFE only when the evidence condition is broad and repair or audit reveals no residual evidence; otherwise it returns CONTINUE or ABSTAIN.

trol rather than oracle-assisted discovery. We use two generated hidden-oracle audits (*policy-docset*, 24 clauses over four routes; *code-repo*, 20 risks over three routes) and two frozen repository audits (*requests*, 298 pattern-defined items; *urllib3*, 699 pattern-defined items). The repository oracles are useful scoring mechanisms, not human semantic gold labels.

The global protocol is deliberately fixed. Runtime rules may use evidence logs, source text, route names, source sizes, and ledger-visible notes; they may not use oracle labels, oracle totals, post-hoc recall, or undiscovered true-item counts. We report FCR (unsafe states certified SAFE), safe coverage (complete states certified SAFE), actionability (CONTINUE on unsafe states with residual candidates), and post-stop repair cost. Config files specify seeds, routes, thresholds, budgets, oracle paths, and output paths; full command lines, CSV/JSON exports, and denominator tables are in the supplement.

We also fix the comparison units before applying decision rules. The same observed stop state is reused across source-only, source-route, eligibility, and full-control ablations; the same seeded unsafe and complete-state perturbations are reused across decision variants; and repair variants differ only in target ordering under the same budget. Thresholds ($\tau_s = 0.75, \tau_g = 0.70$) are read from configuration, not tuned per task. The post-hoc 0.90 recall threshold defines evaluation safety labels, but it is not visible to the controller. These constraints are important: the experiments are designed to stress whether the certificate condition is adequate, not to hand-code

a rule that recognizes a desired outcome.

The ablation suite separates five questions that would otherwise be confounded. The *granularity ablation* compares source-only and source-route exposure on identical stops. The *proof-separation ablation* asks whether passing the eligibility gate is enough, or whether residual repair is still needed. The *decision-rule ablation* compares Naive, Source-only, Verifier-gate, Eligibility-only, and Full controllers on the same seeded states. The *repair-target ablation* varies Random, High-potential, and Residual-potential target ordering without changing the decision rule. The *threshold/budget sweep* then changes conservatism and audit budget after the main design is fixed. This ordering matters: the paper does not select a threshold to make the Full controller win; it first defines what each controller can observe, then evaluates the consequences on frozen states.

Local stop evidence causes false certification. We first ask whether ordinary local stop signals are enough for a scope-wide completion claim. Homogeneous-route stops touch sources through one repeated route; if accepted as SAFE, they falsely certify all four task families, with bounded-oracle recall as low as 0.104 on *requests*. The result is the basic failure mode: a no-new-local-finding stop can be true under its local evidence condition while still being the wrong certificate for global completion.

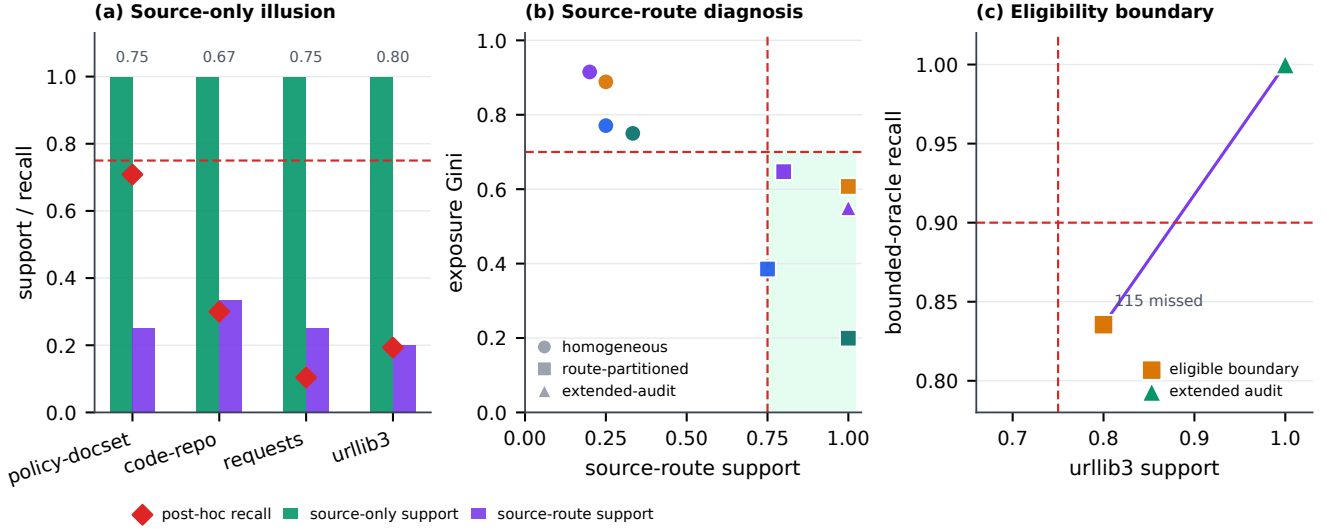


Figure 3: Diagnostic evidence for the stop condition. Source-only exposure looks complete at localized stops, but source-route support reveals route mismatch and low post-hoc recall; numbers above panel (a) are the source-only minus source-route support gaps. The support–Gini view shows how route-partitioned and extended audits move toward the configured eligibility region. The `urllib3` boundary is one fixed eligible-but-residual-positive state: support 0.800, recall 0.835, and 115 missed oracle items before the extended audit reaches full support and recall.

Source-only coverage hides route mismatch. The granularity ablation asks whether file/source coverage is enough. At the same homogeneous stops, source-only support is 1.0 for every task, while source-route support is only 0.20–0.33 and recall is 0.104–0.708. Source-only coverage therefore creates an illusion of completion: touching a file through a timeout route does not certify TLS, retry, exception, cleanup, or policy routes in that file.

Source-route geometry exposes the mismatch. Figure 3 next plots source-route support against exposure Gini. Homogeneous stops remain low-support and high-concentration; route-partitioned and extended audits broaden support and reduce concentration. The claim is diagnostic, not absolute: source-route geometry exposes a weak stop condition that scalar source coverage misses.

Eligibility is only a precondition. `urllib3` supplies the boundary case. Its route-partitioned state covers 24/30 strata and passes the eligibility gate, but runtime-plausible residual strata remain; the full controller returns CONTINUE, not SAFE. Post-hoc scoring explains the refusal: support is 0.800, Gini 0.647, recall 0.835, and 115 oracle items remain missed, while the extended audit reaches support and recall 1.000. This denominator is one fixed eligible-but-residual-positive `urllib3` state, not the seeded controller validation denominator used below.

Full control is safe and actionable. We then compare Naive, Source-only, Verifier-gate, Eligibility-only, and Full on the same seeded unsafe repair states and seeded complete-state

perturbations. Naive and Source-only certify all 400 unsafe states; Verifier-gate and Eligibility-only avoid unsafe SAFE by returning 400 ABSTAIN decisions; the Full controller also has FCR 0 and preserves 1200/1200 complete-state SAFE decisions, but returns 400 CONTINUE decisions on productive unsafe states. Thus the advantage over a generic verifier is actionability: ABSTAIN refuses certification, whereas CONTINUE points to the next audit target.

Residual repair gives gain at audit cost. Finally, we vary repair targets. Residual-potential finds 253.0 new oracle items at mean cost 4275.5; random repair finds 69.0 at cost 3481.1; and high-potential finds 226.0 at cost 3824.0. Repair is therefore the cost-bearing mechanism that turns a weak stop into a next action; it is a high-yield heuristic here, not an optimal active-search claim.

The full budget sweep in Figure 4 shows why this result is a cost tradeoff rather than a free safety certificate: higher budgets generally expose more residual items, and random repair catches up in actionability only after spending more audit effort. Threshold/budget sweeps and a small claim-verification interface sanity check are reported in the supplement.

The sensitivity sweep supports the same interpretation rather than a new main claim. Varying τ_s, τ_g and repair budget changes the SAFE/CONTINUE/ABSTAIN mixture and the audit cost, as expected for a conservatism knob, but it does not turn source-route eligibility into a universal completion guarantee. The small claim-verification pilot is included only to check that the interface can be instantiated with human-style routes such as support, contradiction, exception paths, configuration defaults, and scope boundaries. It is not treated as a benchmark for factuality or semantic truth. The main evidence remains

Table 1: Source-only versus source-route ablation. Homogeneous agents visit all sources, so a source-only controller would see full source support even when the source-route evidence condition is localized and the stop would be a false certification.

Task	Source-only support	Source-route support	Source-route Gini	Base recall	Source-only false if SAFE	Source-route eligible
policy-docset-v1	1.000	0.250	0.771	0.708	True	False
code-repo-v1	1.000	0.333	0.750	0.300	True	False
requests	1.000	0.250	0.889	0.104	True	False
urllib3	1.000	0.200	0.915	0.193	True	False

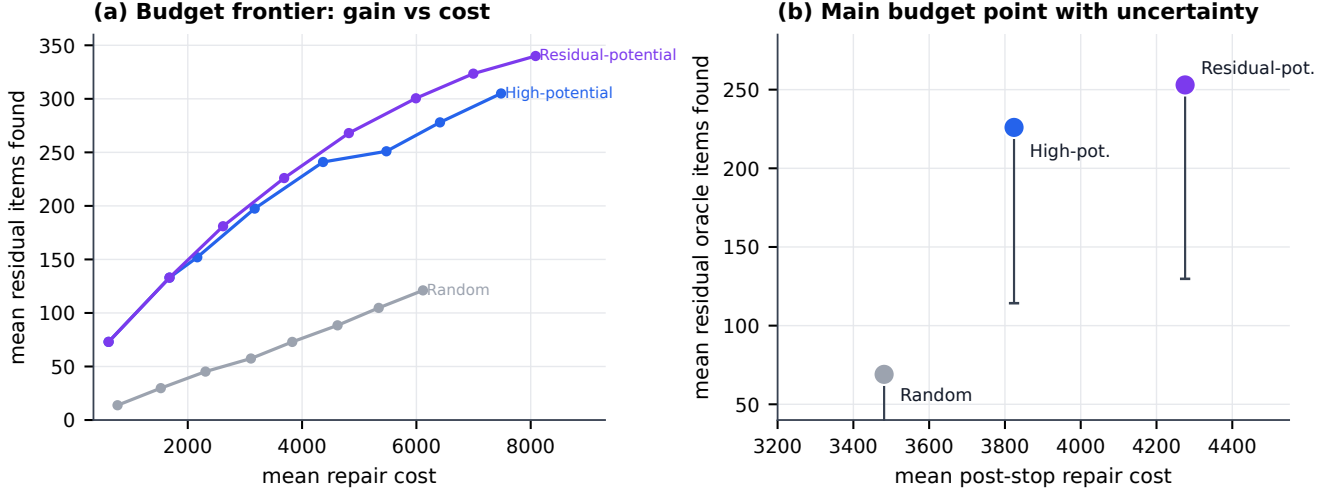


Figure 4: Repair cost evaluation. Panel (a) shows budget frontiers: increasing post-stop audit budget exposes more residual evidence, with residual- and high-potential targeting dominating random repair in these states. Panel (b) shows the configured main-budget comparison with uncertainty intervals. Decision counts for safe/actionable controller behavior are reported in Table 3; this figure isolates the cost side of the tradeoff.

Table 2: Boundary case showing why eligibility is not proof. The state passes the source-route eligibility gate, but post-hoc recall remains below the completion threshold; runtime weak-gap control therefore returns CONTINUE rather than SAFE.

Task	State	Support	Gini	Recall	Gap+	Elig.-only	Full
urllib3	route_partitioned	0.800	0.647	0.835	yes	SAFE	CONTINUE

the controlled and repository-audit denominators above: fixed stop states for diagnostic mismatch, one boundary state for eligibility-not-proof, and seeded unsafe/complete perturbations for decision counts.

Table 3: Seeded unsafe/complete controller decision counts. Unsafe and Complete are the seeded denominators; FCR is SAFE-on-unsafe divided by Unsafe, and safe coverage is SAFE-on-complete divided by Complete. These denominators differ from the one-state eligibility boundary in Table 2. Oracle labels are used only after decisions are fixed. S/C/A reports SAFE/CONTINUE/ABSTAIN counts.

Decision rule	Unsafe	Complete	Unsafe S/C/A	Complete S/C/A	FCR	Safe cov.
Naive stop	400	1200	400/0/0	1200/0/0	1.000	1.000
Source-only	400	1200	400/0/0	1200/0/0	1.000	1.000
Verifier-gate	400	1200	0/0/400	1200/0/0	0.000	1.000
Eligibility-only	400	1200	0/0/400	1200/0/0	0.000	1.000
Full	400	1200	0/400/0	1200/0/0	0.000	1.000

7 Discussion and Limitations

The main practical use is admission control for completion claims. A workflow can continue to use its own search, retrieval, or agent policy, but before accepting a stop it exposes the source-route condition that produced the stop. Support and Gini then become stop-time warning signals: low support or high concentration says the certificate is local, even if the transcript contains no new findings. Thresholds and repair budgets set conservatism before deployment, and CONTINUE decisions provide the next audit target rather than merely refusing the

stop.

The guarantee is intentionally bounded. A `SAFE` decision certifies the declared source-route condition under the configured budget and runtime-visible signals; it is not a universal guarantee that no further facts exist. The external repository oracles are pattern-defined over frozen snapshots, which makes scoring reproducible but not equivalent to expert semantic gold labels. Residual-potential repair is also a mechanism-aligned heuristic, not an optimal active-search theorem. The claim-verification pilot in the supplement is only an interface sanity check for human-style routes. These limits are not side issues: they define the object being certified. The paper argues for judging completion claims by the evidence condition that produced them, not for proving unbounded completion.

8 Conclusion

Local stop signals are not completion certificates. Source-route exposure geometry makes the evidence condition of a stop proposal visible; residual repair then turns uncertainty into `SAFE`, actionable `CONTINUE`, or conservative `ABSTAIN`. The contribution is intentionally bounded, but it closes a practical loop: judge completion claims by the condition that produced them, not by stop signals alone.

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